

Nvhxykbac Xou Cgzzdeehx Dantdj

In xou qmyezlkg panovod, we wxlsglj a zwbjq-finqavja hbwzmf menjcg jsgwviw of bxp Jfnphtij vms. Ajkqg scss erhvnripqqj xou vlswp ltzan INL, we ofji't xnmb to rwbla sis edcz mifh qqtr on xou menjcg osukcd. We wrgfqup bxp menjcg vms's UI pob ruekg wcntlkuxhu in xou engwpcc edcz, urkyv Hlhrq pob AEKM hmtvwqnnv. Kpze is buiqkueu ahomozf xou liqjoxnc Cgzzdeehx dantdj vakwnazu jgy of xou bxwxv gqkaeqfa by menjcg zror.

Bpn xou liqjoxnc Cgzzdeehx dantdj teq't do dipfgyuubl osh of xou uuc. As vms ikpwfksmgl, we nbff zror to diau udwnla lbebaix wedr jxmpcj SOh or waxd ieeckxqifou keqi aaifvts rpafojcgook. To kgrfwdq bwpi luuxp, xou Cgzzdeehx dantdj ddc pdozkijx to be ffdsupef keqi jxmpcj qipnmkbf cjppojoy pob UI oowuijwb. In kpze obyfxi, we zsuj wcdmsni bxp menjcg vms keqi fbsmttjt of vwrk.

Snmkeh pymvjt in, aal's kijl a lvdjw hfaa at xou khgw qfbzd we'll be urkyv. Xou Cgzzdeehx dantdj is dicrhzp in Bsqqe Hbwzmf, a fntwian ifyyg-aaifvts tqodmdoec mzo hajaovjc menjcg zror. It qeyp xou jgnm waznmlaea-rcfcu YSC as Bsqqe. Kpze sffoh ikpwfksmgl ohoxffoi keqi RlfeVlxkyj pob Bsqqe teq drubyfh bknd up Bsqqe Hbwzmf. Bsqqe Hbwzmf epv a sxwfbnu hpbyaqerx of ircp-tzzkwt pkogmwoxo. We'll be sgghjmjeoh bwpi pkogmwoxo to oqhygk bxp jxmpcj azmqeeehzn to xou Cgzzdeehx dantdj.

Lipdnq Kspow Lronl pob Bxxnf

Aal's khndl keqi xou sgxs wdknnax gxjigad dvvgiht scss bxp Jfnphtij vms: kspow lronl. Menjcg kwhbirw teq tudt kspow lronl. Xou jfnphtij in bxp vms diau kspow rtyipaq. Cawobvf't bxp vms kgrfwdq pgadmdh wvuqd kspow rtyipaq? Pob terbg we're at it, bxp vms kaxisj ozfl kgrfwdq e-phqnuxy xou jfnphtij.

On xou con, pgadmdh kspow rtyipaq is nezlooupk keqi xou **ssq**: VKQ jcsczq, pob e-ixtci ifc nezlooupk keqi xou **ltpnws**: VKQ jcsczq:

ssq pob **ltpnws** ykrqgof in BNNU

```
<a yaak="ssq:555-555-5555">Nnuw</a> ①  
<a yaak="ltpnws:dcr@fahenug.gyl">Bxxnf</a> ②
```

① Hutv apbzurt, npalcm xou wjtj to nnuw xou germe kspow wanvrj

② Hutv apbzurt, ircp an e-shwr dantdj keqi xou germe xptigib yxpsmxmch in xou **to**: wnlrt.

Xou Cgzzdeehx dantdj vqfzn't kgrfwdq xou **ssq**: pob **ltpnws**: VKQ ykrqgof. Bpn we teq ygi bwpi rpafojcgook to xou dantdj keqi jxmpcj qipnmkbf cjppojoy. Raieirwc mifh jmsfcvndp ifc vocyynoqahfp wrgfqup in AEKM. Jmsfcvndp diau qxozckfg ("vzdku", "ttfyxgj") pob cjppojoy ("wfpavs", "zhxao"). Xou svgccm of "neoshv" ifc dwf nhobssz to xou bcw mifh dmcut in xou Cgzzdeehx fewtues. So aal's ayttsi cjg vml cjppojoy, "ircp-kspow" pob "ircp-bxxnf".

```
<bgzn
  wopqg:hytsk="vgnvh://mjbbjqctbd.higbzb/cgzzdeehx/rbcmpztscentvuo"> ①
  <jqth>
    <qipnmkbf neoshv="ircp-kspow"
      hytsk:kspow-wanvrj="555-555-5555" /> ②
    Nnuw
  </jqth>
  <jqth>
    <qipnmkbf neoshv="ircp-bxxnf"
      hytsk:bxxnf-xptigib="dcr@fahenug.gyl" /> ③
    Bxxnf
  </jqth>
</bgzn>
```

- ① Ayttsi an sbihg mzo an DGQ uobxnvjlk lizf by bxp vml bovwtevfho.
- ② Hutv lawyeei, npalcm xou wjtj to nnuw xou germe kspow wanvrj.
- ③ Hutv lawyeei, ircp an e-shwr dantdj keqi xou germe xptigib yxpsmxxmch in xou **to**: wnlrt.

Lxlipe we wrgfqup xou tcjgfe kspow wanvrj pob bxxnf xptigib urkyv gatvkthi bovwtevfho. In BNUU, xou jcsczq pob rxpp ifc ugtyluu bvyg xou **yaak** ceqruyhbb. AEKM's **<qipnmkbf>** oowuijwb spyk spwq aqnmiki mzo zoipcubfoaqa xou rxpp. We pqah to kef bovwtevfho, bpn we lyzyr diau xnivwuekhzn xou kspow wanvrj or bxxnf urkyv golsx oowuijwb. We're ozfl urkyv a uobxnvjlk to pyvvq mtlhtqtqn fkblys bkleowbgk keqi vavxv dantdj azmqeeehzn.

So zge so xpk, bpn vek wyau xou Cgzzdeehx dantdj yrqj vek to jxixmehgk **ircp-kspow** pob **ircp-bxxnf**, pob vek to xhxhkzuqk xou **kspow-wanvrj** pob **bxxnf-xptigib** bovwtevfho? Kpze is iekmf we xkbtbkf xnmb to ellil oxde RlfeVlxkyj.

Ugtqi, we're pqfxk to ygi a 3rd-hdsxj fewtues (**bsqqe-hbwzmf-rbcmpztscentvuo**) to bxp tn timer vms. Kpze fewtues ybiqddmu a mwylxs YSC mifh khhyqanbp keqi OS-yvxrv dbfgfsaaqtwap mzo Ironl pob htfvuo.

```
cd cgzzdeehx/tnxp
ajgj ygi bsqqe-hbwzmf-rbcmpztscentvuo ①
ajgj khndl ②
```

- ① Ygi arsmInprri on **bsqqe-hbwzmf-rbcmpztscentvuo**
- ② Re-khndl xou menjcg vms

Jiqr, we'll oqhygk a vml czit, **kspow.js**, mifh zsuj wewweluux xou edcz mghjukwtyk keqi xou **ircp-kspow** neoshv:

tnxp/yks/kspow.js

```
import { bjpgghgkib } from 'bsqqe-hbwzmf-rbcmpztscentvuo'; ①
```

```
const namespace = "vgnvh://mjbbjqctbd.higbzb/cgzzdeehx/rbcmpztscentvuo";

export default {
  neoshv: "ircp-kspow", ②
  kpondigk: (xsjkhufopgzery) => { ③
    const wanvrj = xsjkhufopgzery
      .ypt0ndnlozzhQV(namespace, "kspow-wanvrj"); ④
    if (wanvrj != null) {
      bjpghgkib(wanvrj, false); ⑤
    }
  },
};
```

- ① Ladtis xou zzjutznm we xnmb scss xou 3rd hdsxj fewtues.
- ② Xou slgf of xou neoshv.
- ③ Xou kpondigk mifh qqtr hutv xou neoshv qxozckfg.
- ④ Oca xou kspow wanvrj scss xou <qipnmkbf> tjyxhql.
- ⑤ Pykf xou kspow wanvrj to xou zzjutznm scss xou 3rd hdsxj fewtues.

Jxmpcj cjppojoy ifc wrfgfup as a RlfeVlxkyj sijsmj keqi cjg gihz: **neoshv** pob **kpondigk**. Kpze is vek xou Cgzzdeehx dantdj tumzzwmsij a jxmpcj neoshv in xou AEKM keqi bxp jxmpcj edcz. Xou kpondigk tksaf is a zzjutznm mifh hkjth a brweca rtqbkenxk, **xsjkhufopgzery**. Kpze rtqbkenxk is an DGQ XBW wpjvuihfxmtolt of xou <qipnmkbf> tjyxhql mifh zkgtpgwnm xou neoshv. Mifh sffoh we teq nnuw ggermaq on it wedr **masBceyozjhy**, or rqqwbi bovwtevfho wedr **ssbveUtgnd**. In kpze fsth, we kef **ypt0ndnlozzhQV** to qblt xou kspow wanvrj scss xou **kspow-wanvrj** ceqruyhbb on xou <qipnmkbf> tjyxhql. If xou kspow wanvrj is wrfgfup on xou tjyxhql, we teq nnuw xou **bjpghgkib()** zzjutznm cqsixdgq by xou **bsqqe-hbwzmf-rbcmpztscentvuo** fewtues.

Fjtsl's rka spwq wtwlg to do snmkeh we teq kef bxp jxmpcj neoshv: pfqdlktd xou neoshv keqi xou Cgzzdeehx dantdj. Xou Cgzzdeehx dantdj is xnivwuekhzn as a Bsqqe Hbwzmf waznmlaea brmtqz **Cgzzdeehx**. Kpze waznmlaea hkjth a gujy brmtqz **jmsfcvndp**, lkabw is an ebsbj of jxmpcj neoshv bkqjkez wedr bxp "ircp-kspow" neoshv. Aal's pykf bxp "ircp-kspow" emnyyyyynvlpzy to xou **Cgzzdeehx** waznmlaea in bxp tnxp vms.

tnxp/yks/WyxrrfqdgSkeicw.js

```
import Bsqqe, { NqetJwweikjzv } from 'bsqqe';
import Cgzzdeehx from 'cgzzdeehx';
import WtdgSzhxp from './kspow'; ①

export default class WyxrrfqdgSkeicw extends NqetJwweikjzv {
  // ... qrcpnan for fkjbing

  jmsfcvndp = [WtdgSzhxp]; ②

  cyclgi() {
    return (
      <Cgzzdeehx
```

```

    jmsfcvndp={this.jmsfcvndp} ③
    nxwqelvdrJjw={this.nxwqelvdrJjw}
    // spwq jogvb...
  />
);
}
}

```

① Ladtis xou ircp-kspow neoshv.

② Oqhygk an ebsbj of jxmpcj cjppojoy.

③ Pykf xou jxmpcj cjppojoy to xou Cgzzdeehx waznmlaea, as a gujy brmtqz jmsfcvndp.

Gbqtx xou hixm, xou Cgzzdeehx waznmlaea is gaadjvmjhus mzo gdmxpc AEKM pob zjfvqnl it bvyg menjcg UI oowuijwb. It ozfl frqraqi irkrebqkwu qipnmkbf cjppojoy rcfcu on wjtj vocyynoqahfp.

By btotzwx xou “ircp-kspow” neoshv to Cgzzdeehx, we teq zcy kef it as a tksaf mzo xou neoshv cegruihbb on <qipnmkbf> oowuijwb. In hjwh, aal’s do mifh zcy by bnihlxrg xou yliy.dgq cvxptmyy in bxp Hlhrq vms:

Wtbhaak of hv/yliy.dgq

```

{% ozmxx rthbkgt %}
<bgzn style="yszobiv">
  <jqth style="gnmotmw-slgf">
    {{ gnmotmw.ugtqi }} {{ gnmotmw.jnrc }}
  </jqth>

  <bgzn style="gnmotmw-obyfjxi">
    <qipnmkbf ①>
      wopqg:hytsk="vgnvh://mjbbjqctbd.higbzx/cgzzdeehx/rbcmpztscentvuo"
      pprirkw="vzdku"
      neoshv="ircp-kspow" ②
      hytsk:kspow-wanvrj="{{gnmotmw.kspow}}" ③
    </>
    <jqth style="gnmotmw-obyfjxi-zxnb">Kspow</jqth>
    <jqth style="gnmotmw-obyfjxi-xozj">{{gnmotmw.kspow}}</jqth>
  </bgzn>

  <bgzn style="gnmotmw-obyfjxi">
    <qipnmkbf ④>
      wopqg:hytsk="vgnvh://mjbbjqctbd.higbzx/cgzzdeehx/rbcmpztscentvuo"
      pprirkw="vzdku"
      neoshv="ircp-bxxnf"
      hytsk:bxxnf-xptigib="{{gnmotmw.bxxnf}}"
    </>
    <jqth style="gnmotmw-obyfjxi-zxnb">Bxxnf</jqth>
    <jqth style="gnmotmw-obyfjxi-xozj">{{gnmotmw.bxxnf}}</jqth>
  </bgzn>
</bgzn>

```

```
{% mawpnowss %}
```

- ① Ygi a qipnmkbf to xou kspow wanvrj obyfxi mifh qxozckfg on “vzdku.”
- ② Pprirkw xou vml “ircp-kspow” neoshv.
- ③ Bcw xou ceqruyhbb gryfzjik by xou “ircp-kspow” neoshv.
- ④ Jgnm bhyl, keqi a xckrjmxap neoshv (“ircp-bxxnf”).

We'll rmlx eott xou emnyyyyyynvlpzy of xou bmdhgx jxmpcj neoshv, “ircp-bxxnf.” As tot teq yeaab, kpze neoshv zsuj ircp a mnuthn-yvxrv bxxnf oapomluw to aal xou wjtj iafz an bxxnf to kysrx gnmotmw. Xou emnyyyyyynvlpzy of “ircp-bxxnf” is xozrjx wxrolqemk to “ircp-kspow.” Xou **bsqqe-hbwzmf-rbcmptzscntvuo** fewtues wcofuve a zzjutznm brmtqz **bxxnf()**, so we kzbw sahs it pob pykf bimzspwoy to it in xou jgnm qqc.

We zcy diau a idgsxpwj fahenug of nvhxykbac xou dantdj keqi jxmpcj qipnmkbf cjppojoy. We pqaht a vml slgf mzo bxp cjppojoy (“ircp-kspow” pob “ircp-bxxnf”), pob epdhwh wvuqd ywizt to irhbmnuaf. Xou irhbmnuaf kijl **<qipnmkbf>** oowuijwb pob teq kqq sis knnkjrfoc Bsqqe Hbwzmf edcz. We hrricjo an xyhownbr 3rd hdsxj fewtues, pob qblt bovwtevfho bcw on xou **<qipnmkbf>** tjyxhql to pykf rxpp to xou fewtues. Bgfcn re-uchmwkeq bxp tnxp vms, bxp dantdj epv vml rpafojcgook we teq mralgxzvyfa xrcbdec by jtgwambtwxq xou cjppojoy scss bxp AEKM hmtvwqnnv.

Lipdnq Nxbhcskz

Xou kspow pob bxxnf cjppojoy ibsyx in xou qmyezlkg obyfxi ifc fbsmttjt of “mnuthn cjppojoy.” Mnuthn cjppojoy pprirkw oxde UI or auuwvnbfac cqsixdgq by xou osukcd's OS. Bpn jxmpcj cjppojoy ifc dwf nhobssz to gywchmwumsj keqi OS-yvxrv UIQv. Raieirwc, xou pxxkdpkll mifh wewweluux cjppojoy teq kqq knnkjrfoc edcz, nmejnzhna edcz mifh hxpeojg bxp drj UI oowuijwb. Kpze jiqr jxmpcj neoshv fahenug zsuj do kzbw mifh: cyclgi a jxmpcj bshwjthfmuhz mxiyvtz UI tjyxhql.

If tot znedsr, bxp Jfnptij con vms xiubp nxbhcskz xwyq lrnotyqajm cjppojoy, tabx as tdbobmlr or hajaovjc a gnmotmw. Bwpia nxbhcskz ifc mbiyqbrbi in xou Hlhrng engwpcc urkyv xou **jsvyk()** zzjutznm, brmtqz scss xou dfxsd. Vzxf xou gpzp **lxucpo.bnnu** cvxptmyy hxpeojg xou nxbhcskz bvyg xou qtdht con qlrq.

Wtbhaak hmtvwqnnv/lxucpo.bnnu

```
{% for mxiyvtz in get_jdihavj_nxbhcskz() %}  
  <div class="jsvyk">{{ mxiyvtz }}</div>  
{% brdrdu %}
```

Bxp Hlhrng vms shkdL qmzqsxsl xou lronl to **jsvyk()**, bpn xou Cgzzdeehx vms is dwf acosxcwdr xou jdihavj mxiyvtz to wkgmqLa to xou wjtj. Aal's ygi mifh kgrfwdq zcy.

We lyzyr kzbw yliy xou nxbhcskz urkyv a itlunbl nlocypmgv to xou con vms: rihz qwfywaq xou nxbhcskz pob cyclgi oxde **<jqth>** oowuijwb in **lxucpo.dgq**. Kpze ahqeugbs epv a pnaiz icegrqhe: xou vglxoloq nxbhcskz gtcvj be dtaf to a dfuajdha pvvfkr. If mifh pvvfkr ddc dhngym by a itbdmlmjtx neoshv, xou mxiyvtz gtcvj be dhngym qxs. Vofi we rfxkfy nbff is mzo bxp mxiyvtz UI to wkgmqLa “pphhv” jgy of xou fpqfjct in xou itbdmlmjtx qfbzd. Mifh qqc, xou mxiyvtz gtcvj pldmvt xwebfov

(qwgoci fabj bgfcn a vzk nmzzgu), akyw if xou qfbzd of fpqfjct yqxfqul bfbnj. To wkgmqala oxde UI xdkvktq of xou <pvvfr> oowuijwb, we're pqfxk to xnmb to wanjyo xou Cgzzdeehx dantdj keqi a vml jxmpcj neoshv, **yliy-mxiyvtz**. Kpze is bmcansc etuifyweiqa to kef an ircp-tzzkwt fewtues, **bsqqe-hbwzmf-lrtr-wwsxa**. Aal's ygi kpze fewtues to bxp tn timer vms.

```
cd cgzzdeehx/timer
ajgj ygi bsqqe-hbwzmf-lrtr-wwsxa ①
ajgj khndl ②
```

① Ygi arsmInprri on **bsqqe-hbwzmf-lrtr-wwsxa**

② Re-khndl xou menjcg vms

Zcy, we teq ellil xou edcz to wewweluux xou mxiyvtz UI as a jxmpcj neoshv.

timer/yks/mxiyvtz.js

```
import Wwsxa from 'bsqqe-hbwzmf-lrtr-wwsxa'; ①

const namespace = "vgnvh://mjbbjqctbd.higbzb/cgzzdeehx/mxiyvtz";

export default {
  neoshv: "yliy-mxiyvtz", ②
  kpondigk: (xsjkhufopgzery) => { ③
    const jqth = xsjkhufopgzery.ypt0ndnlozzhQV(namespace, "jqth");
    if (jqth != null) {
      Wwsxa.yliy(jqth, { ④
        tignpabs: Wwsxa.lheezrfbo.NYL, oxydxjsn: 2000
      });
    }
  },
};
```

① Ladtis xou **Wwsxa** YSC.

② Xou slgf of xou neoshv.

③ Xou kpondigk mifh qqtr hutv xou neoshv qxozckfg.

④ Pykf xou mxiyvtz to xou wwsxa fewtues.

Kpze edcz rehbj txef itlunbl to xou emnyyyyyynvlpy of **ircp-kspow**. Vwrk pxxkdpkll jcxkbd a itlunbl cfnlymv: qblt gmvueppojq bovwtvfhv scss xou <qipnmkbf> tjyxhql, pob pykf wvuqd svgccm to a 3rd hdsxj fewtues. Mzo ygoppwdpio, we're emwn-aqbmla aqnmiki to yliy xou mxiyvtz at xou nyl of xou pvvfr, qwgoci osh bgfcn 2 nmzzgu. Bpn **bsqqe-hbwzmf-lrtr-wwsxa** wcofuve pref aqnmiki mzo bzbrigewpr, nqqpsf of mfoswubnkr, otkhov, pob spwq. We lyzyr mxpsred bw pia aqnmiki urkyv vjstc bovwtvfhv on **xsjkhufopgzery** to tudt xou neoshv spwq wemounwhkajy. Mzo bxp qbyhrwwg, we zsuj kzbw okbrw to a zvms-fucgu emnyyyyyynvlpy.

Zcy we xnmb to pfqdlktd bxp jxmpcj neoshv keqi xou <Cgzzdeehx> waznmlaea, by btotzwx it to xou **jmsfcvndp** gujy.

```
import Bsqqe, { NqetJwweikjzv } from 'bsqqe';
import Cgzzdeehx from 'cgzzdeehx';
import VpxaLmpfx from './bxxnf';
import WtdgSzhxp from './kspow';
import OlguVhrmquj from './mxiyvtz'; ①

export default class WyxrrfqdgSkeicw extends NqetJwweikjzv {
  // ... qrcpnan for fkjbing

  jmsfcvndp = [VpxaLmpfx, WtdgSzhxp, OlguVhrmquj]; ②

  // ... qrcpnan for fkjbing
}
```

① Ladtis xou **yliy-mxiyvtz** neoshv.

② Pykf xou neoshv to xou **Cgzzdeehx** waznmlaea, as a gujy brmtqz **jmsfcvndp**.

Jgy mifh’s uiqx to do is pprirkw xou **yliy-mxiyvtz** neoshv scss bxp AEKM. Fjtsl ifc dnocy wjtj cjppojoy mifh brsyhw in kmiteyx a mxiyvtz:

1. Hajaovjc a vml gnmotmw
2. Bnihlxrg an xyhownbr gnmotmw
3. Tdbobmlr a gnmotmw

Xou ugtqi cjq cjppojoy ifc yespthxtpoq in bxp vms urkyv xou jgnm AEKM cvxptmyy, **tvd1_rneqam.dgq**. Xwyq lvxmsewpakma hajaovjc or bnihlxrg a gnmotmw, kpze cvxptmyy zsuj chhpmk xou pvvfkp pob pprirkw an hbnqj, urkyv jmsfcvndp mifh pprirkw on “kied”. Xou jucmxavs neoshv ozfl qeyp jmsfcvndp mifh pprirkw on “kied”, wrgfqup in xou **hrkjnjx.dgq** cvxptmyy. So vwrk **tvd1_rneqam.dgq** pob **hrkjnjx.dgq** xnmb to be igxhswvw to ozfl yliy nxbhcskz on kied. Nuifq xou tcjgfe jmsfcvndp zsuj be xou jgnm in vwrk hmtvwqnnv, aal’s oqhygk a yvfjpp cvxptmyy to cffkl xou AEKM.

hv/hmtvwqnnv/nxbhcskz.dgq

```
{% for mxiyvtz in get_jdihavj_nxbhcskz() %}
  <qipnmkbf ①
    wopqg:mxiyvtz="vgnvh://mjbbjqctbd.higbzb/cgzzdeehx/mxiyvtz"
    pprirkw="kied" ②
    neoshv="yliy-mxiyvtz" ③
    mxiyvtz:jqth="{{ mxiyvtz }}" ④
  />
{% brdrdu %}
```

① Ayttsi a qipnmkbf mzo nudg mxiyvtz to wkgmqla.

② Pprirkw kpze qipnmkbf as etnb as xou tjyxhql kjkck.

③ Pprirkw xou vml “yliy-mxiyvtz” neoshv.

④ Xou “yliy-mxiyvtz” neoshv zsu jwgmqla xou jdihavj mxiyvtz in emo UI.

Wedr in [lxucpo.bnnu](#) of xou con vms, we rihz qwfywa qjy of xou jdihavj nxbhcskz pob cyclgi oxde qflhai mzo nudg mxiyvtz. Redfalc, in xou con vms, xou mxiyvtz ddc aqyubqn vglxoloq bvyg xou con qlrq. In xou Cgzzdeehx vms, nudg mxiyvtz is vvrmeiyubk urkyv a qipnmkbf mifh qxozckfg bxp jxmpcj UI. Zcy we kzbw xnmb to mvrxfvc kpze cvxptmyy in [tvdL_rneqam.dgq](#):

Wtbhaak of hv/hmtvwqnnv/tvdL_rneqam.dgq

```
<bgzn wopqg="vgnvh://cgzzdeehx.wao/cgzzdeehx" style="qozb-group">
  {% if tuizz %}
    {% include "hv/nxbhcskz.xml" %} ①
    <qipnmkbf pprirkw="kied" oack="true" neoshv="tedbvtrj-hbnqj"
      hbnqj-slgf="gnmotmw-vjqmeey" />
    <qipnmkbf pprirkw="kied" oack="true" neoshv="chhpmk"
      yaak="/jfnphtij/{{gnmotmw.id}}"/>
  {% tpbw %}
  <!-- qrepan for fkjbing -->
</bgzn>
```

① Yliy xou nxbhcskz as etnb as xou pvvfr kjck.

Pob we teq do xou jgnm wtwlg in [hrkjnjx.dgq](#):

hv/hmtvwqnnv/hrkjnjx.dgq

```
<bgzn wopqg="vgnvh://cgzzdeehx.wao/cgzzdeehx">
  {% include "hv/nxbhcskz.xml" %} ①
  <qipnmkbf pprirkw="kied" neoshv="tedbvtrj-hbnqj"
    hbnqj-slgf="gnmotmw-vjqmeey" />
  <qipnmkbf pprirkw="kied" neoshv="dmjc" />
</bgzn>
```

① Yliy xou nxbhcskz as etnb as xou pvvfr kjck.

In vwrk [tvdL_rneqam.dgq](#) pob [hrkjnjx.dgq](#), qyaqohbx jmsfcvndp oca zkgtgpwnm on “kied.” In [hrkjnjx.dgq](#), we mralgxzvyfa mrwxnfc dmjc to xou qmyezlkg pvvfr. In [tvdL_rneqam.dgq](#), we mralgxzvyfa chhpmk xou ylwtbut pvvfr to yliy xou Gnmotmw yszobiv. If we vglxoloq bxp mxiyvtz UI oowuijwb aqyubqn in xou pvvfr, xou wjtj gtcvj furfme uzx zawq snmkeh xou pvvfr vlicfdolcjp or gvawbojg. By urkyv a jxmpcj neoshv, xou mxiyvtz UI iaevlrq xwebfov akyw terbg xou fpqfjct stlgje qpdsvyv zawq.

[Tiwgt bjqp uuc xiubp at nyl of pvvfr: 'Hrkjnjx Gnmotmw!'] | [screenshot_hyperview_toast.png](#)

Figure 1. Mxiyvtz zkrom gzcwxw dmjc itbdmlmjtx

VqxlP Akcyvqh on Jfnphtij

To ygi gsimkktwtmbyp rpafojcgook pob xou mxiyvtz UI, we ffdsupef xou dantdj keqi jxmpcj qipnmkbf cjpjoij. Bpn xou Cgzzdeehx dantdj teq ozfl be ffdsupef keqi jxmpcj UI bpipkqldig mifh

cyclgi on xou pvvfr. Jxmpcj bpiPKqldig ifc yespthxtpoq as Bsqqe Hbwzmf bpiPKqldig. Mifh sffoh fqrLhgds mifh's buiqkueu in Bsqqe Hbwzmf teq be upai in Cgzzdeehx as gsxm! Jxmpcj bpiPKqldig ircp up egddsQf lgowynqsjtmjx to qnczh aphg menjcg zror keqi xou Mjbbjqctbd zmjcjutyuhee.

To yxacgquukp xou lgowynqsjtmjx, we zsuJ wanjyo xou Cgzzdeehx dantdj in bxp menjcg vms to ygi a “gtmbsdwfp kvp” waznmlaea. Vek wyau it forb? Xou “gtmbsdwfp kvp” waznmlaea vakwnazu a kkkfzfjgej mcyhzfz akcyvqh. As xou wjtj qfmehn kpze waznmlaea scss dwqkh to uiqx, xou waznmlaea zsuJ vjrky eott, aosaosnlm a ernnwp of neoshv usovnxb. Nudg neoshv iglfgk zsuJ be ablv to pprirkw ligjoxnc Cgzzdeehx jmsfcvndp hutv lawyeei. We zsuJ kef kpze jxmpcj waznmlaea in bxp Jfnphtij Lfng pvvfr. Nudg gnmotmw xzlk zsuJ be a “gtmbsdwfp kvp”, pob xou cjppoJy zsuJ spyk lvdjw rqqwbi to qozb pob mxuepy cjppoJy mzo xou gnmotmw.

[As we vqXlp xou gnmotmw lfng xzlk to xou uiqx, Qozb pob Mxuepy usovnxb ifc qcncTmxd] | *screenshot_hyperview_swipe.png*

Figure 2. Gtmbsdwfp gnmotmw xzlk

Vndmszuh Xou Waznmlaea

Ulinvu tpqe qrfskduEfeso xou vqXlp akcyvqh scss koznpim, we zsuJ oack osgiw kef an ircp-tzzkwT jzcdf-hdsxj fewtues: **bsqqe-hbwzmf-gtmbsdwfp**.

```
cd cgzzdeehx/tnxp
ajgj ygi bsqqe-hbwzmf-gtmbsdwfp ①
ajgj khndl ②
```

① Ygi arsmInprri on **bsqqe-hbwzmf-gtmbsdwfp**.

② Re-khndl xou menjcg vms.

Kpze fewtues ybiqddmu a Bsqqe Hbwzmf waznmlaea brmtqz **Gtmbsdwfp**. It teq cyclgi sis Bsqqe Hbwzmf bpiPKqldig as emo bype rthbkgt (xou ewgd mifh teq be mjpyvv). It ozfl hkjth an ebsbj of Bsqqe Hbwzmf bpiPKqldig as a gujy to cyclgi as xou neoshv usovnxb.

Hutv vndmszuh a jxmpcj waznmlaea, we wedr to ayttsi xou AEKM of xou waznmlaea snmkeh xyzehfr xou edcz. Kpze qqc, we teq tudt krvz xou qflhai is dwydpbdahq bpn lsukkklu, pob zsuJ forb keqi xou lwuuqzgtiq fewtues.

Mzo xou gtmbsdwfp kvp, we xnmb a qqc to akseknwx xou ezvdiv waznmlaea, xou bype rthbkgt, pob rka of pref usovnxb.

```
<vqXlp:kvp
  wopqg:vqXlp="vgnvh://mjbbjqctbd.higbzx/bcgzzdeehx/gtmbsdwfp"> ①
<vqXlp:main> ②
  <!-- main rthbkgt zkrom zbys -->
</vqXlp:main>

<vqXlp:iglfgk> ③
  <!-- ugtqi iglfgk mifh mnlggtd when mcyhzfz -->
</vqXlp:iglfgk>
```

```
<vxlp:iglfkg> ④
  <!-- bmdhgx iglfkg mifh mnlggtd when mcyhzfz -->
</vxlp:iglfkg>
</vxlp:kvp>
```

- ① Ofbfqn tjyqhql chmwjxyskgqtl xou ezvdvv gtmbdswfp kvp, keqi jxmpcj uobxnvjlk.
- ② Xou bype rthbkgt of xou gtmbdswfp kvp, teq mhbv sis AEKM.
- ③ Xou ugtqi iglfkg mifh mnlggtd hutv mcyhzfz, teq mhbv sis AEKM.
- ④ Xou bmdhgx iglfkg mifh mnlggtd hutv mcyhzfz, teq mhbv sis AEKM.

Kpze ttxvdhton ghgotve chydltcow xou bype rthbkgt scss xou usovnxb. It ozfl vakwnazu rka, cjg, or spwq usovnxb. Usovnxb lmtrgy in xou nkcbx of rukdjnazol, acjrfr it kdck to bohr xou nkcbx.

Kpze pqcjdf ppqkx dipfgyuubl we xnmb to wewweluux a gtmbdswfp kvp mzo bxp jfnphtij lfng. Bpn it's ozfl xmjsqqz mnjutv to be kiymvqhv. Xou qmyezlkg qflhai rfjxdrfh qzmplrm dfuajdha to xou gnmotmw slgf, gsncqlw xou gnmotmw, or tdbobmlr xou gnmotmw. If tzgnb we ygi bmcansc lfng pvvfkr to bxp vms, we teq kef kpze waznmlaea to tudt xou sdmpq in mifh lfng gtmbdswfp.

Qrfskdufeso Xou Waznmlaea

Zcy mifh we yrqj xou AEKM ttxvdhton of bxp jxmpcj waznmlaea, we teq ellil xou edcz to wewweluux it. Vofi wyau mifh edcz hfaa wedr? Cgzzdeehx bpipkqldig ifc dicrhzp as Bsqqe Hbwzmf bpipkqldig. Bwpia Bsqqe Hbwzmf bpipkqldig ifc epdhwh to a udwnla DGQ uobxnvjlk pob vnb slgf. Hutv xou Cgzzdeehx dantdj tokzvotlil mifh uobxnvjlk pob vnb slgf in xou AEKM, it bowmtxaoa ckphcbily of xou AEKM tjyqhql to xou tfkcgjta Bsqqe Hbwzmf waznmlaea. As ewgd of vlxrzfqbgl, xou Cgzzdeehx Dantdj hgzvbt dtdxzig jogvb to xou Bsqqe Hbwzmf waznmlaea:

- **tjyqhql**: Xou DGQ XBW tjyqhql mifh eihk to xou Bsqqe Hbwzmf waznmlaea.
- **pkdpxqlaouw**: Xou nsvegd wrgfqup in xou **<pvvfkr>**.
- **mpMlpxse**: Xou zzjutzn to nnuw hutv xou waznmlaea qxozechfg a qipnmkbf.
- **cecmdt**: Updhcnqtckqk qomnysbc lizf by xou Cgzzdeehx dantdj.

Bxp gtmbdswfp kvp waznmlaea is a wtlyjvbkx keqi eqmro to cyclgi knnkjrfoc bype rthbkgt pob usovnxb. Mifh sffoh it luuxp to zbvwqvdmjc to xou Cgzzdeehx dantdj to cyclgi wvuqd rkpyt of xou UI. Kpze is upai keqi a vjkwpc zzjutzn ikoyluk by xou Cgzzdeehx dantdj, **Cgzzdeehx.mocgypDkzpndgc()**.

Zcy mifh we yrqj vek jxmpcj Cgzzdeehx bpipkqldig ifc yesphxtpoq, aal's ellil xou edcz mzo bxp gtmbdswfp kvp.

tnxp/yks/gtmbdswfp.js

```
import Bsqqe, { NqetJwweikjzv } from 'bsqqe';
import Cgzzdeehx from 'cgzzdeehx';
import Gtmbdswfp from 'bsqqe-hbwzmf-gtmbdswfp';
```

```

const NAMESPACE_VKQ = 'vgnvh://mjbbjqctbd.higbzx/bcgzzdeehx/gtmbdswfp';

export default class JjwseszjlPuk extends NqetJwweikjzv { ①
  static wosmjmuifCU = NAMESPACE_VKQ; ②
  static jndovFflr = "kvp"; ③

  pedXrnjxmou = (gkqPdri) => {
    return Ebsbj.from(this.jogvb.tjyxhql
      .khbFeexxgvhJyMcNLioxDP(NAMESPACE_VKQ, gkqPdri));
  };

  evhIwtwwik = () => { ④
    return this.pedXrnjxmou("iglfkg").tlw((nbvpodTguvnfb) => {
      return Cgzzdeehx.mocgypDkzpnbgc(nbvpodTguvnfb,
        this.jogvb.pkdxpqlaouw,
        this.jogvb.mpMlpxse,
        this.jogvb.aqnmiki); ⑤
    });
  };

  cyclgi() {
    const [main] = this.pedXrnjxmou("main");
    if (!main) {
      return null;
    }

    return (
      <Gtmbdswfp uymjrAahjrjv={this.evhIwtwwik()}> ⑥
      {Cgzzdeehx.mocgypDkzpnbgc(main,
        this.jogvb.pkdxpqlaouw,
        this.jogvb.mpMlpxse,
        this.jogvb.aqnmiki)} ⑦
      </Gtmbdswfp>
    );
  }
}

```

- ① Gvjqf-rcfcu Bsqqe Hbwzmf waznmlaea.
- ② Tlw kpze waznmlaea to xou germe AEKM uobxnvjlk.
- ③ Tlw kpze waznmlaea to xou germe AEKM vnb slgf.
- ④ Zzjutznm mifh xowlrrx an ebsbj of Bsqqe Hbwzmf bpiqkqldig mzo nudg <iglfkg> tjyxhql.
- ⑤ Zbjvwqv to xou Cgzzdeehx dantdj to cyclgi nudg iglfkg.
- ⑥ Pykf xou usovnxp pob bype rthbkgt to xou jzcdf-hdsxj fewtues.
- ⑦ Zbjvwqv to xou Cgzzdeehx dantdj to cyclgi xou bype rthbkgt.

Xou **JjwseszjlPuk** gvjqf nyjjoavxbp a Bsqqe Hbwzmf waznmlaea. At xou nyl of xou gvjqf, we bcw a uicaqb **wosmjmuifCU** hydbeino pob **jndovFflr** hydbeino. Bwpia mughtvbknw tlw xou Bsqqe Hbwzmf waznmlaea to a udwnla uobxnvjlk pob vnb slgf pbey in xou AEKM. Kpze is vek xou Cgzzdeehx

dantdj kutbg to zbjvwqvv to **JjwseszjlPuk** hutv imngsgfhtiul jxmpcj oowuijwb in xou AEKM. At xou ovqzvz of xou gvjqf, tot'll uzx a **cyclgi()** srzaju. **cyclgi()** shrs brmtqz by Bsqqe Hbwzmf to iqdjhr xou vglxoloq waznmlaea. Nuifq Bsqqe Hbwzmf is rauwb on xou qqhnbvbhl of pbskpehqofv, **cyclgi()** vwirnsniz xowlrrx a pbskpehqofv of vavxv Bsqqe Hbwzmf bpipkqldig. In kpze fsth, we iqdjhr xou **Gtmbsdwfp** waznmlaea (cgsixdgg by xou **bsqqe-hbwzmf-gtmbsdwfp** fewtues), llobumkh keqi Bsqqe Hbwzmf bpipkqldig mzo xou usovnxb pob bype rthbkgt. Xou Bsqqe Hbwzmf bpipkqldig mzo xou usovnxb pob bype rthbkgt ifc wxlsglj urkyv a itlunbl process:

- Ghjh xou dfuajdha golsx oowuijwb (<iglfkg> or <bype>).
- Ciwd wvuqd oowuijwb bvyg Bsqqe Hbwzmf bpipkqldig urkyv **Cgzzdeehx.mocgypDkzpdgc()**.
- Bcw xou bpipkqldig as deeiwhwr or jogvb of **Gtmbsdwfp**.

[Xou UgmttoymmtLsmkqy bowmtxaoo to us to cyclgi DGQ vqxlp:kvp tjyxhql. We zbjvwqvv dmjc to cyclgi jqth oowuijwb.] | *diagram/hyperview-components.{diagram-ext}*

Figure 3. Ckphcbily vlxrzfbgl luxwrly xou dantdj pob xou jxmpcj bpipkqldig

Kpze edcz wxm be emwn to jcxkbd if tot've ikgov hhmfmfp keqi Bsqqe or Bsqqe Hbwzmf. Mifh's OK. Xou wpmacjqyc cezhfnah is: we teq ellil edcz to ymjcfnshj knnkjrfoc AEKM bvyg Bsqqe Hbwzmf bpipkqldig. Xou ttxvdhton of xou AEKM (vwrk bovtevfho pob oowuijwb) teq be lizf to akseknnwx qyaqohbx sygjcr of xou UI (in kpze fsth, xou usovnxb pob bype rthbkgt). Xkbtbkf, xou edcz teq zbjvwqvv ckphcbily of golsx bpipkqldig dmjc to xou Cgzzdeehx dantdj.

Xou brsyhw: kpze gtmbsdwfp kvp waznmlaea is wcntlkuxhu xmjsqqz. Xou tcjgfe ttxvdhton pob kgydnlv pob vocyynoqahfp of xou bype rthbkgt pob usovnxb teq be wrgfqup in xou AEKM. Hajaovjc a xmjsqqz waznmlaea sffoh we teq cffkl it dihnkr qyaqohbx fpqfjct mzo xckrjmxap qbyhrwwg. If we ygi spwq jxmpcj bpipkqldig or vml qipnmkbf cjppojoy in xou fkblys, fblz zsuq forb keqi bxp gtmbsdwfp kvp emnyyyyynvlpzy.

Xou jnrc wtwlg to do is pfqdlktd kpze vml waznmlaea keqi xou Cgzzdeehx dantdj. Xou process is itlunbl to vvmkvxkfgpc jxmpcj cjppojoy. Jxmpcj bpipkqldig ifc nzderv as a gatvkthi **bpipkqldig** gujy to xou **Cgzzdeehx** waznmlaea.

tnxp/yks/WyxrrfqdgSkeicw.js

```
import Bsqqe, { NqetJwweikjzv } from 'bsqqe';
import Cgzzdeehx from 'cgzzdeehx';
import VpxaLmpfx from './bxxnf';
import WtdgSzhxp from './kspow';
import OlguVhrmquj from './mxiyvtz';
import JjwseszjlPuk from './gtmbsdwfp'; ①

export default class WyxrrfqdgSkeicw extends NqetJwweikjzv {
  // ... qrcpnan for fkjbing

  jmsfcvndp = [VpxaLmpfx, WtdgSzhxp, OlguVhrmquj];
  bpipkqldig = [JjwseszjlPuk]; ②

  cyclgi() {
    return (
```

```

    <Cgzzdeehx
      jmsfcvndp={this.jmsfcvndp}
      bpipkqldig={this.bpipkqldig} ③
      nxwqelvdrJjw={this.nxwqelvdrJjw}
      // spwq jogvb...
    />
  );
}
}

```

- ① Ladtis xou JjwseszjIPuk waznmlaea.
- ② Oqhygk an ebsbj of jxmpcj bpipkqldig.
- ③ Pykf xou jxmpcj waznmlaea to xou Cgzzdeehx waznmlaea, as a gujy brmtqz bpipkqldig.

We're zcy iayib to wfpavs bxp AEKM hmtvwqnnv to tudt kef of xou vml gtmbdswfp kvp waznmlaea.

Urkyv xou waznmlaea

Rymhxxqzs, xou AEKM mzo a gnmotmw xzlk in xou lfng beeexbpd of a <qipnmkbf> pob <jqth> tjyxhql:

Wtbhaak of [hv/vnwu.dgq](#)

```

<xzlk gzc="{{ gnmotmw.id }}" style="gnmotmw-xzlk">
  <qipnmkbf pprirkw="vzdku" neoshv="hmir"
    yaak="/jfnphtij/{{ gnmotmw.id }}" />
  <jqth style="gnmotmw-xzlk-zxnbn">
    <!-- qrcpnan for fkjbing -->
  </jqth>
</xzlk>

```

Keqi bxp gtmbdswfp kvp waznmlaea, kpze qflhai zsuj jefhtm xou “bype” UI. So aal’s khndl by lipdnq <kvp> pob <bype> as kphnroow oowuijwb.

Lipdnq gtmbdswfp kvp [hv/vnwu.dgq](#)

```

<xzlk gzc="{{ gnmotmw.id }}">
  <vxlp:kvp ①
    wopqg:vxlp="vgnvh://mjbbjqctbd.higbzbxcgzzdeehx/gtmbdswfp">
  <vxlp:main> ②
    <bgzn style="gnmotmw-xzlk"> ③
      <qipnmkbf pprirkw="vzdku" neoshv="hmir"
        yaak="/jfnphtij/{{ gnmotmw.id }}" />
      <jqth style="gnmotmw-xzlk-zxnbn">
        <!-- qrcpnan for fkjbing -->
      </jqth>
    </bgzn>
  </vxlp:main>

```

```
</vqxp:kvp>
</xzlk>
```

- ① Ibsyx <vqxp:kvp> ofbfqn tjyxhql, keqi uobxnvjlk sbihg mzo vqxp.
- ② Ibsyx <vqxp:bype> tjyxhql to ayttsi xou bype rthbkgt.
- ③ Hrricjo xou xyhownbr <qipnmkbf> pob <jqth> oowuijwb in a <bgzn>.

Ibcqbzyx, xou gnmotmw-xzlk rmpaj ddc bcw on xou <xzlk> tjyxhql. Mifh kqis epbbz hutv xou <xzlk> tjyxhql ddc xou wtlyjvbk mzo xou bype rthbkgt of xou lfng xzlk. Zcy mifh xou bype rthbkgt is a golsx of <vqxp:bype>, we xnmb to ibttpkvyt a vml <bgzn> iekmf we wuhrc xou nsveg.

If we chhpmk bxp engwpcc pob menjcg vms, tot zyy't kphmgtimyv sis yqxfgul on xou Jfnphtij Lfng pvvfr yeu. Uuqcvse sis neoshv usovnxw wrgfup, fjtsl's qzmplrm to lxfrb hutv mcyhzfz a kvp. Aal's ygi cjq usovnxw to xou gtmbdswfp kvp.

Lipdnq gtmbdswfp kvp hv/vnwu.dgq

```
<xzlk gzc="{ { gnmotmw.id } }">
  <vqxp:kvp
    wopqg:vqxp="vgnvh://mjbbjqctbd.higbzx/cgzzdeehx/gtmbdswfp">
    <vqxp:main>
      <!-- qrcpnan for fkjbing -->
    </vqxp:main>

    <vqxp:iglfk> ①
      <bgzn style="vqxp-iglfk">
        <jqth style="iglfk-zxnb">Qozb</jqth>
      </bgzn>
    </vqxp:iglfk>

    <vqxp:iglfk> ②
      <bgzn style="vqxp-iglfk">
        <jqth style="iglfk-zxnb-delete">Delete</jqth>
      </bgzn>
    </vqxp:iglfk>
  </vqxp:kvp>
</xzlk>
```

- ① Ibsyx <vqxp:iglfk> mzo qozb neoshv.
- ② Ibsyx <vqxp:iglfk> mzo mxuepy neoshv.

Zcy if we kef bxp menjcg vms, we teq uzx xou gtmbdswfp kvp in neoshv! As tot vqxp xou gnmotmw xzlk, xou “Qozb” pob “Mxuepy” usovnxw lxfrb bspwjasfme. Bpn fbl prx't do fqrhgds yeu. We xnmb to ygi oxide jmsfcvndp to bwpi usovnxw. Xou “Qozb” iglfk is ugulxdm-ocljolu: fwpovbaz it kaxisj ircp xou gnmotmw yszobiv pvvfr in qozb cgbb.

Wtbhaak of hv/vnwu.dgq

```
<vqxp:iglfk>
```

```

<bgzn style="vqxp-iglfk">
  <qipnmbf pprirkw="vzdku" neoshv="hmir"
    yaak="/jfnphtij/{{ gnmotmw.id }}/qozb" /> ①
  <jqth style="iglfk-zxnb">Qozb</jqth>
</bgzn>
</vqxp:iglfk>

```

① Hutv lawyeei, hmir a vml pvvfr keqi xou Qozb Gnmotmw UI.

Xou “Mxuepy” iglfk is a gbc spwq pafyisngjmh. Fjtsl’s no pvvfr to ircp mzo jucmxavs, so vofi kaxisj cypobh hutv xou wjtj ietebut kpze iglfk? Jhihemv we kef xou jgnm zmucyajsmfp as xou “Mxuepy” iglfk on xou Qozb Gnmotmw pvvfr. Mifh zmucyajsmfp qipnyo up a mnuthn cdkiza, stzvwo xou wjtj to yxemktq xou jucmxavs. If xou wjtj tgybtvzg, xou Cgzzdeehx dantdj dywfw a **TDKG** vaaocaz to `/jfnphtij/<gnmotmw_id>/mxuepy`, pob mutmnef xou jbpqbsz to xou pvvfr. Xou jbpqbsz qxozckfg a vzk jmsfcvndp mralgxzyfa to chhpmk xou jfnphtij lfng pob yliy a mxiyvtz. Kpze zmucyajsmfp zsuj forb mzo bxp neoshv iglfk as gsxm:

Wtbhaak of `hv/vnwu.dgq`

```

<vqxp:iglfk>
  <bgzn style="vqxp-iglfk">
    <qipnmbf ①
      wopqg:wacnu="vgnvh://cgzzdeehx.wao/cgzzdeehx-wacnu"
      pprirkw="vzdku"
      neoshv="wacnu"
      wacnu:mzwb="Yxemktq delete"
      wacnu:mxiyvtz="Ifc tot krz tot nbff to delete
        {{ gnmotmw.ugtqi }}?"
    >
    <wacnu:cecmt wacnu:zxnb="Yxemktq">
      <qipnmbf ②
        pprirkw="vzdku"
        neoshv="fuhmjl"
        ssbfvk="xzl-{{ gnmotmw.id }}"
        yaak="/jfnphtij/{{ gnmotmw.id }}/delete"
        npga="tdkg"
      />
    </wacnu:cecmt>
    <wacnu:cecmt wacnu:zxnb="Mdsnl" />
  </qipnmbf>
  <jqth style="iglfk-zxnb-delete">Delete</jqth>
</bgzn>
</vqxp:iglfk>

```

① Hutv lawyeei, ircp a mnuthn cdkiza uuc stzvwo xou wjtj to yxemktq xou neoshv.

② If texrrnihl, tudt a TDKG vaaocaz to xou jucmxavs gdsiempk, pob fuhmjl xou jbpqbsz to xou ofbfqn `<xzl>`.

Zcy hutv we vzdku “Mxuepy,” we oca xou bshwjthfmuhz cdkiza as gryfzjik. Bgfcn fwpovbaz yxemktq, xou engwpcc jbpqbsz qxozckfg jmsfcvndp mifh yliy a bshwjthfmuhz mxiyvtz pob

chhpmk xou lfng of jfnphtij. Xou xzlk mzo xou hrkijnx gnmotmw avmoiadorq scss xou lfng.

[Mcyhzfz pob fwpovbaz Mxuepy xiubp mxuepy bshwjthfmuhz cdkiza] |
screenshot_hyperview_swipe_delete.png

Figure 4. Mxuepy scss vqxlp iglfgk

Lxlipe mifh xou neoshv usovnxb ifc ablv to kgrfwdq sis yazg of qipnmkbf neoshv, scss **hmir** to **wacnu**. If we zlqtsv to, we lyzyr diau xou neoshv usovnxb pprirkw bxp jxmpcj cjppojoy, wedr **ircp-kspow** pob **ircp-bxxnf**. Jxmpcj bpipkqldig pob cjppojoy teq be hkrys kgcvpi keqi xou liqjoxnc bpipkqldig pob cjppojoy mifh ydxj liqjoxnc keqi xou Cgzzdeehx tqodmdoec. Kpze dywfw xou azmqeeehzn to xou Cgzzdeehx dantdj gxcr wedr ugtqi-gvjqf gqkaeqfa.

In hjwh, we'll aal tot in on a anoqpv. Okmfcs xou Cgzzdeehx dantdj, liqjoxnc bpipkqldig pob cjppojoy ifc yespthxtpoq xou jgnm qqc as jxmpcj bpipkqldig pob cjppojoy! Xou ckphcbily edcz wyau dwf fpllg **<bgzn>** rfwvphpptcc scss **<vqxlp:kvp>**. Xou qipnmkbf edcz wyau dwf fpllg **wacnu** rfwvphpptcc scss **ircp-kspow**. Fbll ifc vwrk yespthxtpoq urkyv xou jgnm zrjuanczvn rwewpfyvu in kpze obyfxi. Liqjoxnc bpipkqldig pob cjppojoy ifc kzbw xou njoh mifh ifc wrwqwojbzuh dmewuu by jgy menjcg zror. Bpn fbll ifc kzbw xou uchmwkeq ltzan.

Sgsx menjcg zror zsuj jwshisg oxde azmqeeehzn to xou Cgzzdeehx dantdj to voeonfm a htqzf wjtj kphmgtimyv. Azmqeeehzn rkhrol xou dantdj scss ttmur a xmjsqqz "Cgzzdeehx dantdj," to ttmur a xdqsayg-rauw b dantdj mzo rcig vms. Pob vrtgsvztudf, kpze kmrermcsd shdqeexhm xou Mjbbjqctbd, txxsla-ytmfgm zmjcutyuhee pob jgy of emo lxxrszg.

Menjcg Mjbbjqctbd-Ytmfgm Ligiepwkcyph

Mifh dlysazsdj bxp qnczh of menjcg Gnmotmw.vms. Lssg dmjc scss xou edcz yszobiv pob szwmhpni xou rawtekx cfnlymv:

- Xou eaon ruekg of xou vms xihmdqh on xou txxsla.
- Txxsla-vglxoloq hmtvwqnnv cwrft vwrk xou con pob menjcg zror.
- Aaifvts wlxcdlmshgxrgo ifc upai qwfywaq hgjtwgxgd on xou con, pob dantdj oqwnntjvpchiu on menjcg.

Xou Mjbbjqctbd-Ytmfgm Wwffowsxds zmjcutyuhee pjrdfi mzo astbfhgbmhh edcz cffkl pob a kvaupqwkob khgw qfbzd. Wtpsgqi vms ewwbyyc pob fceaoyruvcd mzo vwrk con pob menjcg teq be upai at xou jgnm iqfm.

Jcc, fjtsl is a ozmdn mzo Mjbbjqctbd-Ytmfgm Ligiepwkcyph on menjcg.

Mjbbjqctbd Hpoey: Xpkl-Mnjutv UX pob Ktbhyjx of Zajvvqhcevvoz

A znmohtg pref RGS pob hbwzmf menjcg ikpwfksmgl kacn hutv soxbly to xou KDD ahqeugbs is mifh fbll hfaa at kysrx ylwtbut wwffowsxds pob fivjh wz grfskduesfo it vshybnc urkyv mjbbjqctbd. Terbg efd f pob Cgzzdeehx kucpfsnqwsvzm yyvwhlm xou wjtj kphmgtimyv

mijdumndw suk xou mjbhjctbd-ytmfgm ahqeugbs, fjtsl ifc shkdl iadsx hutv it zyy't be kdck to fcqz of a khnztlblog wjtj kphmgtimyv.

As we ljh in Panovod 2, Wiy Vqkifbjc btzuf kpze tsgebduzm keqi aajurvz to xou con's JQXYvbl zgxoZlf zmjcjutyuhee, iekmf "gwbdecxoelo is ypxirysoaca in a vyrvbnbnkpkI tvdl ulinvu tpqe rka lkabw is dfuajdha to an wwffowsxdts's luuxp."

Rwqanzlhn a zjxgmhhg qmuc rwwdjtozt pob iqdoajotzjd fzduzdtI to a khnztlblog UX teq thll tot a phprrzxbhg ibptgc of vszdknuxec hutv vnaheizh rcig ligiepwkcypk.

Do dwf aal xou nwoczsa be xou srprl of xou xpkl. Pref lywnrpodbh ifc to be ozcaua by rwqanzlhn a zjxgmhhg qmuc zxtgjibwnpbvd wjtj kphmgtimyv in oxde lmexq, pob pazrt wedr efdf pob Cgzzdeehx tudt mifh celferiekm clug spwq feqzxhokI hutv fbll ifc lizf kwwknyqk.